

Rachel Baker-Ramos

Research Scientist & CS PhD Candidate

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RESEARCH INTERESTS

I'm a Research Scientist in **educational technology** and **community-based design**, with a focus on the intersections of culturally revitalizing education, AI for low-resource languages, and game-based learning. I bring robust experience in human-centered design, participatory research, and project leadership, with **two rigorous design degrees** and **6+ years co-designing with communities**. My work is informed by my multidisciplinary background and Latine heritage, aiming to provide equitable, engaging, and culturally affirming educational experiences to K-12 students.

I currently lead the **Education and Public Interest HCI** research sub-group of Georgia Tech's Ka Moamoa Ubiquitous Computing Lab, under the guidance of Dr. Josiah Hester. I was awarded the **College of Computing's Outstanding Research Scientist Award** for 2025.

WORK EXPERIENCE

- 2024– **Research Scientist**, *College of Computing, Georgia Institute of Technology*
★ College of Computing's 2025 Outstanding Research Scientist Award
Lead the Ka Moamoa Lab's Education & Public Interest HCI sub-group, conducting research--resulting in 8+ peer-reviewed publications. Managing 11+ MS/BS students across 7+ projects. Write grant and award proposals, securing 475k funding from Google Research, Mozilla Foundation, and the NSF.
- 2024– **UX Design & Research Advisor**, *Ulu Lāhui Foundation*
Provide UX design & research support for educational & social impact technology projects. Prepare & write grant proposals, contributing to the successful securing of over 125k funding from MIT Solve and aiEDU.
- 2023-24 **Graduate Research Assistant**, *College of Computing, Georgia Institute of Technology*
- 2023 **UX Design Fellow**, *U.S.C.I.S. Refugee, Asylum & International Operations Directorate*
- 2020-22 **Lead Graphic Designer**, *Veggie Mijas (Latinx/e Environmental Justice Org)*
- 2019-22 **Project Designer**, *Caddis Collaborative (Architecture & Urban Design Studio)*

EDUCATION

- 2026– **Georgia Institute of Technology**, *PhD Computer Science*
Advisors: Josiah Hester (Georgia Tech), Alex Cabral (M.I.T.)
- 2022-24 **Georgia Institute of Technology**, *MS Human-Computer Interaction*
Thesis: *Kumu Connect: Co-designing Culturally-Relevant Generative EdTech with Hawaiian Immersion Educators*,
Advisors: Josiah Hester, Judith Uchidiuno
★ Best MS-HCI Thesis Award Finalist
- 2014-19 **Carnegie Mellon University**, *Bachelor of Architecture (BArch), additional major in Social & Political History*
★ University Honors and College of Fine Arts Honors
★ Senior Leadership Award '2019

PUBLICATIONS

- IDC 2026 **Kumu Connect: Culturally-Authentic AI Tools to Enable Educator Agency and Data Sovereignty for Hawaiian K-12 CS Education** [↗](#)
★ Best Paper Honorable Mention
Rachel Baker-Ramos, Manas Mhasakar, Viswak Raja, Evyn-Bree Helekahi-Kaiwi, William Gelder, Benjamin Carter, Rebecca Diego, Alex Cabral, Josiah Hester. 25th Annual ACM Interaction Design and Children Conference (IDC '26). (Acceptance rate: 23%)
- CHI 2026 **Noondawind: Co-Designed Dashboard for Indigenous Data Access and Environmental Policy Implementation** [↗](#)
★ Best Paper Honorable Mention
Julia Aileen McKenna, Gabriela Buraglia, Jahanvi Kolakaluri, **Rachel Baker-Ramos**, Samantha Carter, Joe Graveen, Jonathan Gilbert, James Rasmussen, Brandon Byrne, Darren Vogt, Josiah Hester, Kim Marion Suiseeya, Alex Cabral. ACM CHI Conference on Human Factors in Computing Systems (CHI '26)
- CHI 2026 **Whose Knowledge Counts? Co-Designing Community-Centered AI Auditing Tools with Educators in Hawai'i** [↗](#)
Dora Zhao, Hannah Cha, Michael J Ryan, Angelina Wang, **Rachel Baker-Ramos**, Evyn-Bree Helekahi-Kaiwi, Rebecca Diego, Josiah Hester, Diyi Yang. ACM CHI Conference on Human Factors in Computing Systems (CHI '26)
- RESPECT 2025 **Kumu Connect: Design Thinking for Place-Based Generative Educational Technology in Hawaiian Immersion Schools** [↗](#)
Rachel Baker-Ramos, Will Gelder, Leah Cho, Jahnavi Kolakaluri, Josiah Hester. 2025 ACM Conference on Research on Equitable and Sustained Participation in Engineering, Computing, and Technology (RESPECT '25).
- RESPECT 2024 **"Those don't work for us": An Assets-Based Approach to Incorporating Emerging Technologies in Viable Hawaiian Teacher Support Tools for Culturally Relevant CS Education** [↗](#)
William Gelder, **Rachel Baker-Ramos**, Ayoung Cho, Jahnavi Kolakaluri, Judith Uchidiuno, Josiah Hester. 2024 ACM Conference on Research on Equitable and Sustained Participation in Engineering, Computing, and Technology (RESPECT '24).
- Koli Calling 2023 **Keeping Mindful of Modality: A Comparison of Computer Science Education Resources for Learning** [↗](#)
Michael J. Johnson, **Rachel Baker-Ramos**, Christopher L. Hovey, Betsy DiSalvo. 23rd ACM Koli Calling International Conference on Computing Education Research (Koli Calling '23)

POSTERS AND ABSTRACTS

- GoodIT 2026 **Still Warm: A Digital Game for Players to Address Climate Grief**
Rachel Baker-Ramos, Josiah Hester, Alex Cabral. ACM 6th International Conference on Information Technology for Social Good (GoodIT26). (Accepted)
- CHI EA 2025 **"I Would Never Trust Anything Western": Kumu (Educator) Perspectives on Use of LLMs for Culturally Revitalizing CS Education in Hawaiian Schools** [↗](#)
Manas Mhasakar*, **Rachel Baker-Ramos***, Ben Carter, Evyn-Bree Helekahi-Kaiwi, Josiah Hester. ACM CHI Conference on Human Factors in Computing Systems (CHI EA '25) *denotes equal contribution
- CHI PLAY 2024 **Enabling Restorative Hawaiian Futurism via a Tabletop Role-Playing Game and Digital Ha'i Mo'olelo (Storytelling)** [↗](#)
Rachel Baker-Ramos, Michael Parkin, Angelica Raza-Furtado, Alika S. Naihe, Jack Hobbs, Jeffrey Vierra, Aarom Ramirez, Chantel Rubin-Molina, Cheradean Kaaialii, Celeste Mason, Yaman Sangar, May Okihiro, Josiah Hester. 2024 ACM Annual Symposium on Computer-Human Interaction in Play (CHI PLAY '24)

AWARDS & FELLOWSHIPS

- 2026 **Best Paper Honorable Mention**, *ACM Interaction Design and Children 25th Conference*
for "Kumu Connect: Culturally-Authentic AI Tools to Enable Educator Agency and Data Sovereignty for Hawaiian K-12 CS Education" (IDC 2026)
- 2025 **CoC Outstanding Research Scientist Award**, *Georgia Institute of Technology, College of Computing (CoC)*
- 2025 **Solver, Indigenous Communities Fellowship**, *MIT Solve*
Kumu Connect was one of 32 solutions selected out of ~2,900 applicants (1% acceptance rate) as part of MIT Solve's Indigenous Communities Fellowship, a program that supports community-based solutions for and with Indigenous communities.
- 2023 **Civic Innovation Fellow**, *Coding It Forward*
Selected from an applicant pool of 2,500 students (5% acceptance rate) as part of the Coding it Forward Summer Fellowship, a program that empowers early-career technologists to explore careers in civic & public interest technology.

SERVICE & VOLUNTEERING

- June 2026 **ACM Interaction Design and Children Conference**, *Student Volunteer*
- 2024 **UX of EdTech**, *Community Volunteer, Summer Cohort*
- 2023-24 **MS-HCI DEI Forum**, *DEI Forum Co-leader*
- 2020-23 **BoulderReads**, *Volunteer ESL Literacy Tutor*

PROJECTS & EFFORTS

- 2026– **Still Warm: A Game on Climate Grief**
- 2025– **AI for Hawai'i (AI4HI)**
Responsibilities: Project Management, Graphic/UX Design
- 2023– **Kumu Connect: A Culturally-Relevant CS Lesson Plan Generator**, *Role: Project Lead*
Responsibilities: Project Management, UX Design, UX Research, Graphic Design
★ 23-24 Best MS-HCI Thesis Award Finalist
★ Google Academic Research Award
★ 2025 MIT Solve Indigenous Communities Fellowship solution
- 2024-25 **Co-Designing AI Bias Auditing with Educators in Hawai'i**, *Role: Community Engagement Lead*
- 2023-26 **Kaona: Educational Role-Playing Game and "Storyteller" Mobile App**, *Role: UX Designer & Game Artist*
Responsibilities: Project Management, UX Design, UX Research, & Graphic Design
- 2023-24 **Internet Outage Detection and Analysis (IODA) Dashboard**, *Role: UX Designer*
- 2023 **Pickrick Project: AR Public History Installation on Atlanta Civil Rights History**, *Role: UX Designer*

GRANTS & FUNDING

- 2026 **NSF Supplement Support Request: Translating Community-Driven Edge AI Data Systems into K-12 STEM Curricula for Rural Classrooms**, *National Science Foundation*
Project: SCC-IRG: Edge AI, Biofabrication, and Data Science for Disaster Preparedness and Community Wellbeing in Hawaii [↗](#)
Role: Personnel & Grant-writer
- 2025 **Indigenous Communities Fellowship**, *MIT Solve*
Project: Kumu Connect
Role: Senior Personnel & Grant-writer

- 2025 **Rural & Indigenous Community Catalyst Program, aiEDU**
Project: AI for Hawai'i Curricula Co-Creation Workshops & Resource Hub
Role: Senior Personnel & Grant-writer
- 2024 **Google Academic Research Award, Google Research**
Project: Kumu Connect: Community-informed AI Teacher Support Tools for Place-Based CS Curriculum in Hawaiian Schools
Role: Senior Personnel & Grant-writer

STUDENT ADVISING & REPORTS

- 2026– **Maha Sidi, MS-HCI, MS Thesis on "Atlanta Beltline & Public History"**
- 2025– **Cheilota Uma, MS-HCI,**
Independent Study on "Kumu Connect: Code Generation" & MS Thesis on "Atlanta Beltline & Public History"
- 2024– **Ben Carter, BS-CS, Independent Study on "PLACES Education", Student Assistantship on "Kumu Connect" & BS Thesis on "Future Hawaiian AI"**
- 2026 **Dharmik, MS-CS, Independent Study on "Kumu Connect: RAG Experiments"**
- 2026 **Hanna Fu, MS-HCI, Independent Study on "Kumu Connect: Code Generation"**
- 2026 **Akanksha Tanwar, MS-CS, Independent Study on "Kumu Connect"**
- 2025-26 **James Kemerait, BS/MS-CS, Research Assistantship on "Kilo Tech"**
- 2025-26 **Isabella Sirianni, MS-HCI, MS Thesis on "Climate Investigators"**
- 2025-26 **Theresa Antony, MS-HCI, Independent Study on "Climate Games" & MS Thesis on "Climate Investigators"**
- 2025 **Viswak Raja, MS-HCI, Graduate Research Assistantship on "Kumu Connect"**
- 2025 **Christine Taylor, MS-HCI, Independent Study on "Noondawind: Bilingual Dashboards"**
- 2024-25 **Manas Mhasakar, MS-CS, Graduate Research Assistantship on "Kumu Connect"**
- 2024-25 **Jingxuan Wang, MS-HCI, Independent Study & Graduate Research Assistantship on "Elections Project"**
- 2024-25 **Jahnvi Kolakaluri, MS-HCI, Independent Study on "Noondawind: Data Storytelling" & Graduate Research Assistantship on "Kumu Connect" and "Noondawind"**
- 2024-25 **Michael Parkin, MS-HCI, Graduate Research Assistantship on "Kaona"**